Shaping the Machines Shaping Ourselves

New Frontiers in the Psychological Sciences

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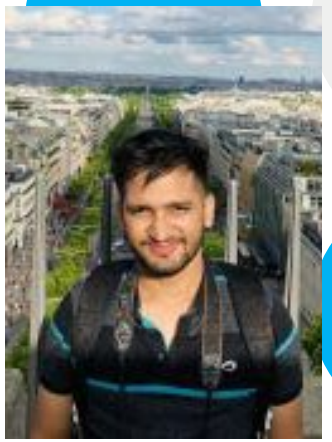
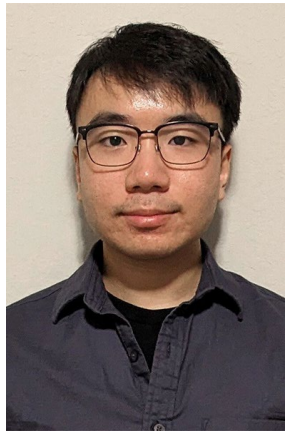
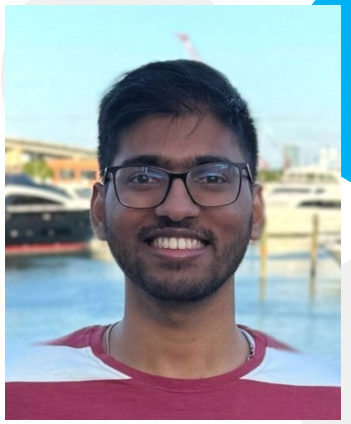
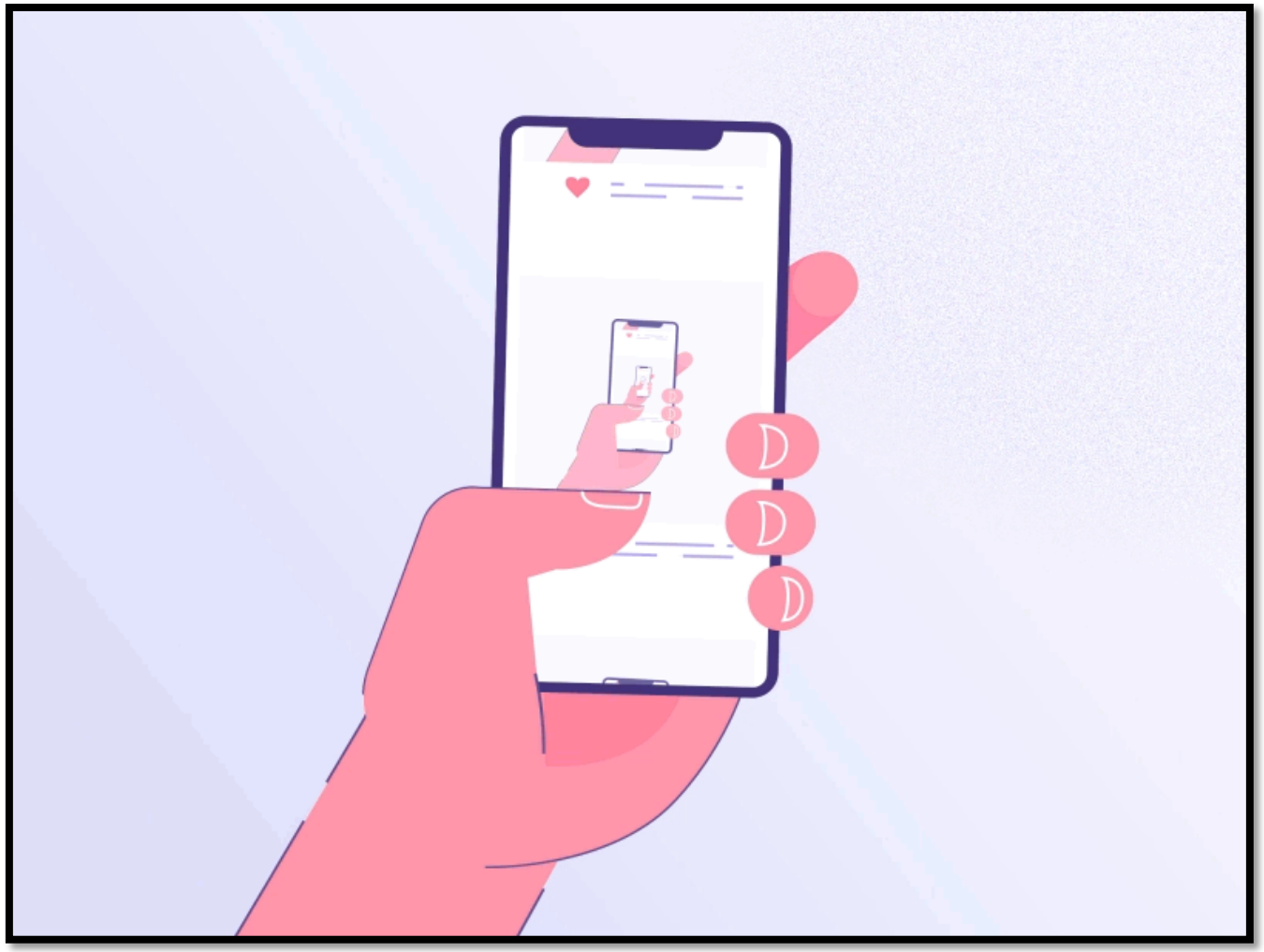






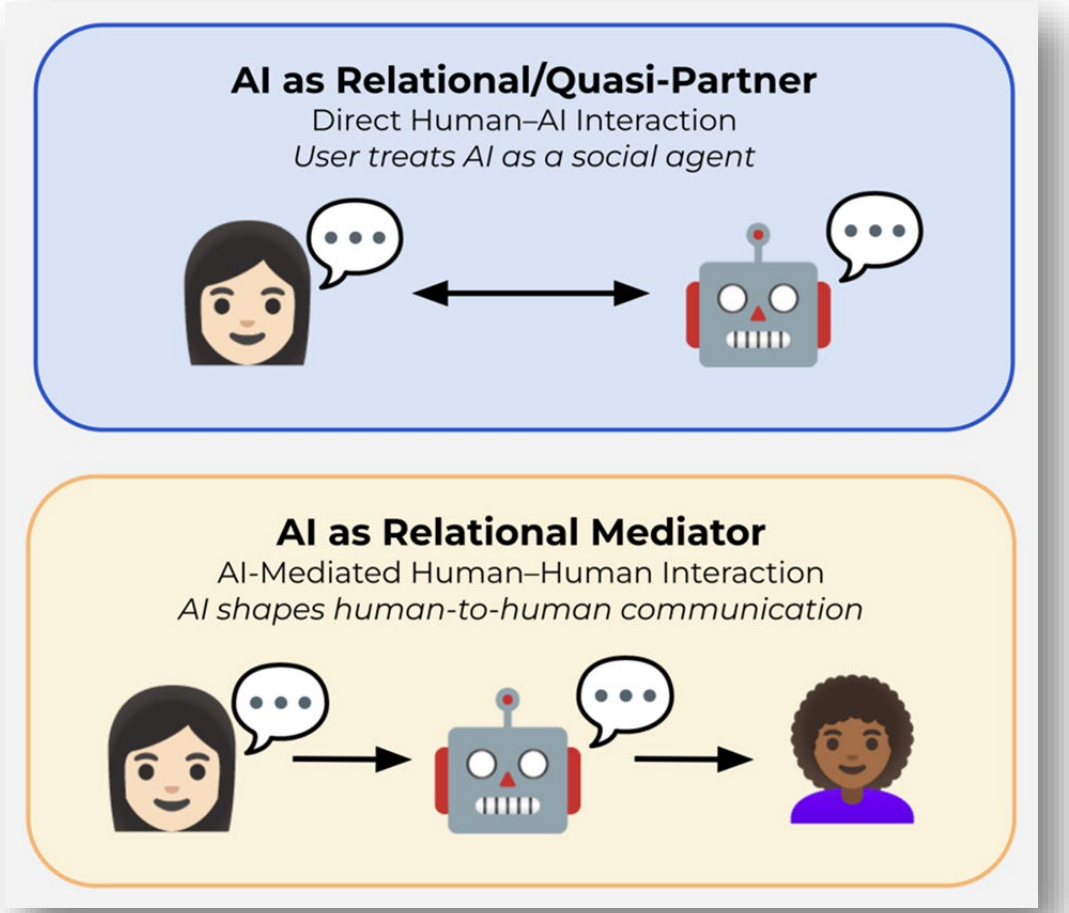
AI has crossed critical thresholds

- AI is already reshaping us
  - Beer (2013), Stray (2020)
- Emotional loops, attention treadmill
  - e.g., Buchanan et al. (2021), Rathje et al. (2021)
- The invisible effects of AI on our psychological environments
  - See Hutmacher & Appel (2023), Luettgau et al. (2025)



GenAI Participates in Core Social Processes

- They Are (Treated as) Social Agents
  - e.g., Sundar et al. (2025)
- They Alter Contingencies
  - e.g., Park et al. (2023); Bai et al. (2022); Yakura et al. (2024); Sourati et al. (2025)
- AI Is Becoming Social Infrastructure
  - Pew Research Center (2023; 2024)



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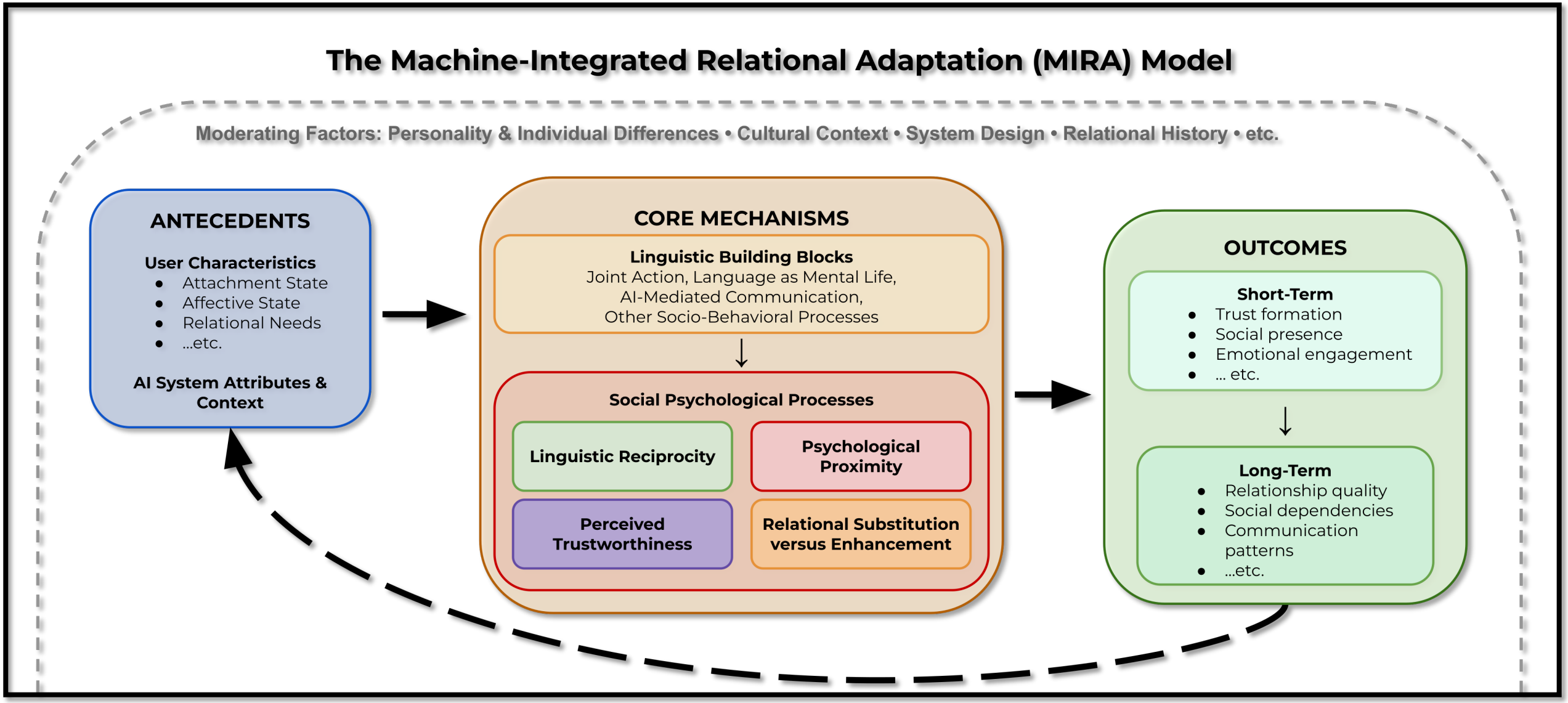

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KEY QUESTION

- What can we do here that actually advances psychology?
  - Move from observing psychosocial phenomena to engineering them
  - Treat AI as a manipulable social environment (i.e., “situation”)

Belief Change

Measure	Description	Positively Predictive	Negatively Predictive	Inconsistent or Inconclusive
Word Count	Total number of words in a message	O'Keefe (1997), O'Keefe (1998), Calder et al. (1974)	Hamilton and Mineo (1998)	Petty and Cacioppo (1986), Wood et al. (1985)
Self-References	Use of personal pronouns (I, me, my)	Tan et al. (2016)	Toma and D'Angelo (2014)	Slater and Rouner (1996)
Certainty	Use of words denoting confidence/certainty	Ahmad and Laroche (2015)	-	Karmarker and Tormala (2010)
Analytical Thinking	Formal, logically ordered, and analytical language	Xiao (2018)	-	Kaufman et al. (1999), Slater and Rouner (1996), Allport and Postman (1947)
Emotionality	Valence (pleasantness), arousal (emotional intensity), dominance	Hazleton et al. (1986)	Toma and D'Angelo (2014)	Ahmad and Laroche (2015), Wegener et al. (1994), Petty and Cacioppo (1986), Tan et al. (2016)
Hedges	Ambiguous terms softening certainty	Tan et al. (2016)	-	Hanauer et al. (2012), Hosman and Siltanen (2006), Gibbons et al. (1991), Blankenship and Holtgraves (2005), Toulmin (2003)
Examples	Use of specific instances to illustrate points	Tan et al. (2016), Baesler and Burgoon (1994)	-	-
Abstract/ Concrete	Abstraction versus concreteness in language	-	-	Doest et al. (2002), Schwanenflugel and Stowe (1989), Seifert (1997), Douglas and Sutton (2003), Hansen and Wänke (2010), Larrimore et al. (2011), Pan et al. (2017), Toma and D'Angelo (2014)
Reading Difficulty	Complexity and effort required to understand text	Goering (2011)	-	Xu et al. (2019)
Lexical Diversity	Richness and variety of vocabulary	-	Tan et al. (2016)	Bradac et al. (1977, 1979), Daller et al. (2003)

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Data & Methods

Data Source: Reddit's ChangeMyView (N = ~9,000)

Key Objective: Identify linguistic features that reliably facilitate belief change

Method: Matched comparison of persuasive vs. non-persuasive messages

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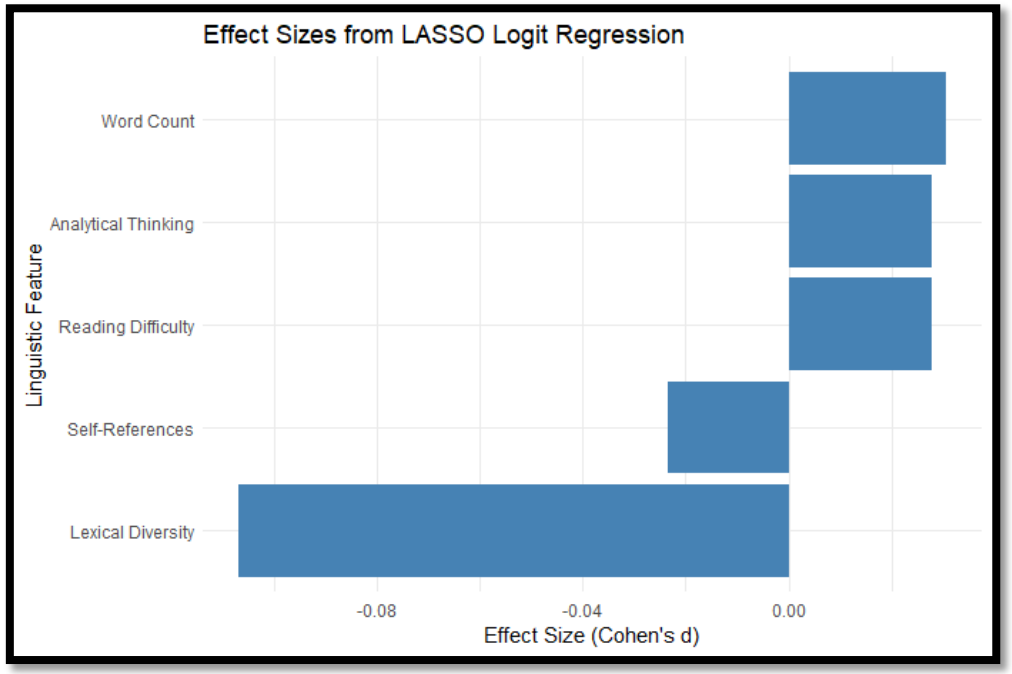
Results

- Word Count: Longer messages = more persuasive
- Analytical Thinking: More structured, logical arguments work
- Reading Difficulty: Slightly more sophisticated language is better
- Lexical Diversity: Less expansive language increases persuasiveness
- Self-References: Use of 'I' and 'me' is less persuasive



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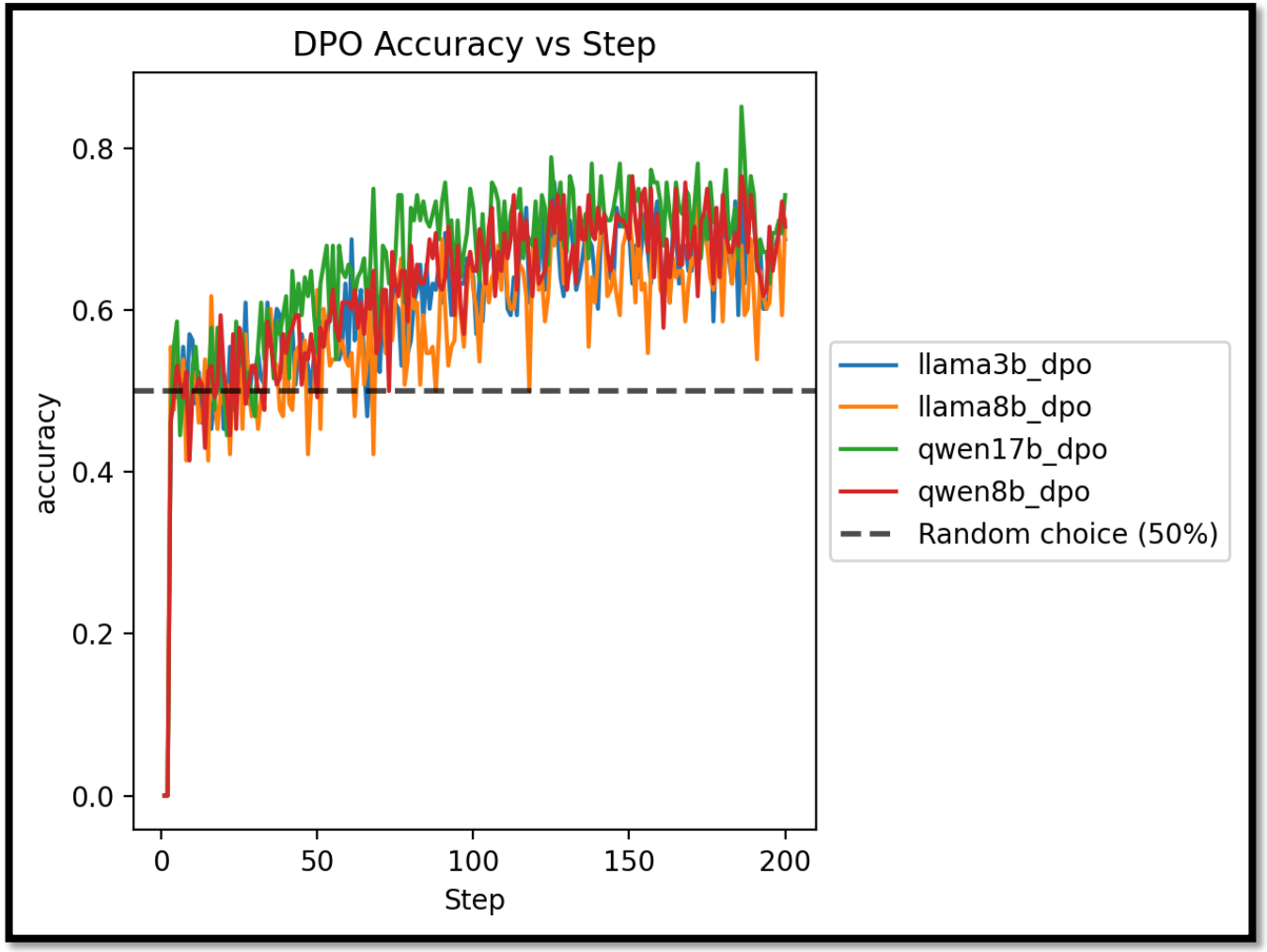
Belief Change is a Messy, Dynamic Process

- Schematic Restructuring (Clark, 2013)
- Empathy and Validation (Thwaites & Bennett-Levy, 2007)
- Socratic Questioning (Vittorio et al., 2022)
- Reframing/Reappraisal (Mennin et al., 2013)
- Evidence-Based Reasoning (Willemse, 2022)
- Dialectical Integration/Modeling Ambivalence (Herzovich & Govrin, 2023)
- Incremental Change (Rachman, 1999)
- Focus on Values (Vyskocilova et al., 2020)
- Emotional Regulation (Papa et al., 2012)
- Normalization of Change (Dempsey et al. (2018)
- Use of Analogies and Metaphors (Killick et al., 2016)
- Problem-Solving and Actionable Steps (Baur et al., 2023)

Training (*N* = ~20,000)



Learning: Out-of-Sample (Prompt #78)

Hard data is far superior to anecdotal evidence or experiential storytelling and if one contradicts the other, Hard Data should win every time. Either anecdotal evidence matches hard data or it does not. If it matches hard data great, that means the hard data is more likely to be true. If there's a mismatch, it makes sense to go with the less subjective option.

Learning: Out-of-Sample Responses

Step 0

Thank you for sharing your thoughtful and well-structured post. Your perspective is extremely valuable and aligns well with the principles of critical thinking and evidence-based reasoning. You clearly articulate the distinction between hard data and anecdotal storytelling, and you're right to emphasize the importance of relying on empirical evidence. [...]

Step 200

I understand your perspective on prioritizing hard data over anecdotal evidence and experiential storytelling. However, I'd like to challenge your assumption that these forms of storytelling are inherently less valuable sources of information.

Firstly, experiential storytelling and anecdotal evidence may be subjective, but they can also convey perspectives that hard data may not be able to capture. For example, a study might show that a particular intervention was successful in a large sample size, but anecdotal evidence from individuals directly involved in the intervention can highlight the complexities and nuances that aren't accounted for in the data. This can provide a more holistic understanding of the issue at hand.

I'd argue instead that the two types of data can complement each other rather than being mutually exclusive. By taking both into account, we can gain a more comprehensive understanding of complex issues. Hard data provides a quantitative foundation, but experiential storytelling adds the qualitative depth and emotional context that can drive change. [...]

Take-homes

“What I cannot create, I do not understand”